Electronics purchase funnel

Where does the observed shopping journey lose momentum? A retrospective analysis of public event data.

Introduction

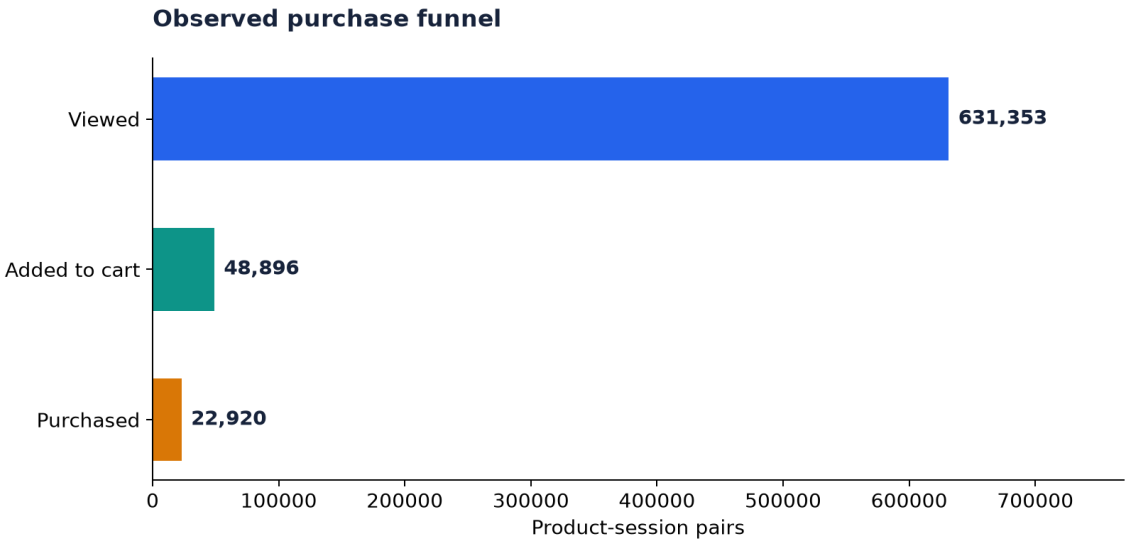
An e-commerce product team needs to decide which part of the shopping journey deserves investigation first. This independent portfolio case study uses public electronics-store events shared by REES46 / Open CDP through Kaggle. Its purpose is to prioritize a product question, not to claim an intervention has improved sales.

The analyzed file contains 885,129 events from 24 September 2020 to 28 February 2021. The actual dates and event types differ from the dataset page description. Only view, cart, and purchase events are present; checkout and cart-removal are not observed.

Problems and measurement

Event counts are not conversion rates: the same shopper may view a product repeatedly or interact with several products in one session. I count distinct user-session-product combinations and require a strict same-product sequence: view < cart < purchase. These figures are not unique people or orders.

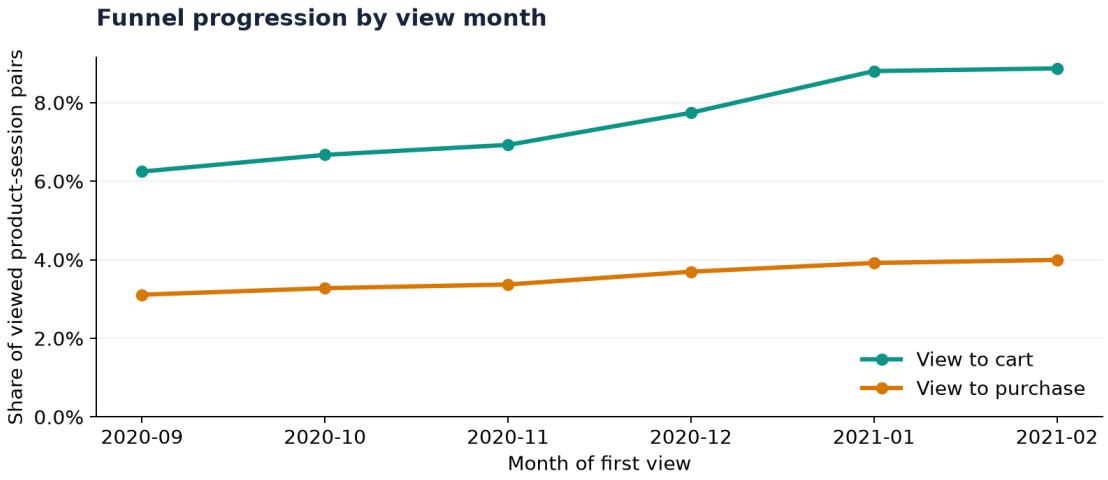
Observed stage	Product-session pairs	Rate
Viewed	631,353	Starting cohort
Added to cart	48,896	7.74% of viewed
Purchased	22,920	46.88% of carted



Strict same-product sequence: view < cart < purchase. Source: REES46 via Kaggle.

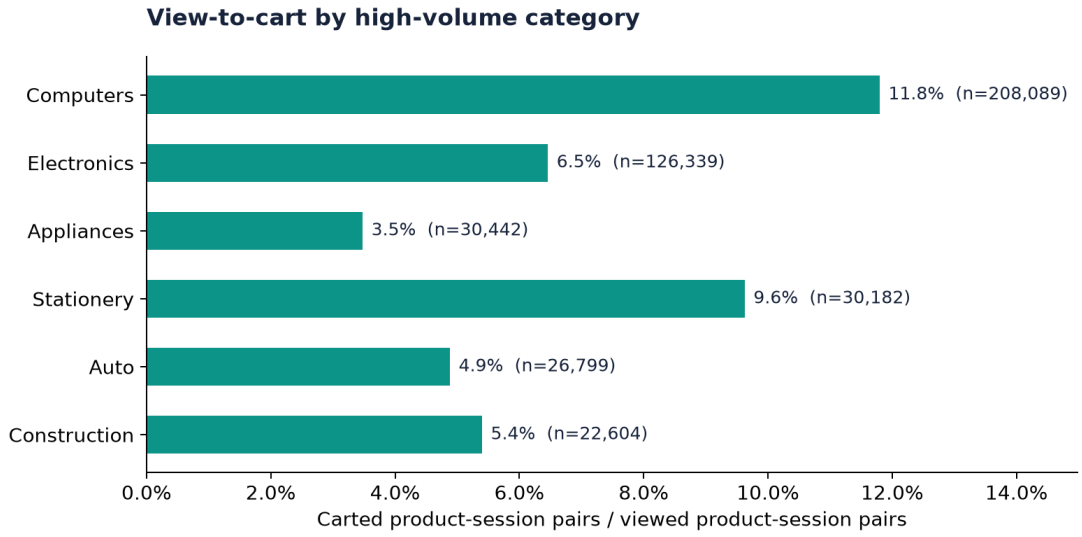
Problems: variation and data coverage

View-to-cart increased from 6.67% for October 2020 view cohorts to 8.81% in January 2021; cart-to-purchase moved from 49.11% to 44.52%. Traffic and product mix can change over time, so this trend is descriptive. September is a partial month, while late-February journeys may continue beyond the file.



September 2020 starts on 24 Sep; February 2021 may have right-edge censoring. Source: REES46 via Kaggle.

Among the largest named categories, view-to-cart ranges from 3.48% for appliances to 11.80% for computers. Category mix, price, availability, and intent may explain the gap. A further 178,822 viewed pairs (28.3%) have no category label; they are not included in the named-category chart.



Six largest named categories. Unknown category excluded (178,822 viewed pairs). Source: REES46 via Kaggle.

Data quality and sensitivity

The funnel excludes 165 events lacking a session token instead of inventing one. The source has 652 repeated event-key rows (same user, session, product, event type, and timestamp); these were not silently removed. There are 145 product-session timestamps with more than one event type. If tied timestamps are allowed as ordered, the purchased stage rises only from 22,920 to 22,943 pairs. The main finding is therefore not sensitive to that timestamp choice.

Among 27,207 product-session pairs with any purchase, 4,287 (15.76%) lack a complete preceding view-and-cart chain in the observed file. This is an observation-coverage problem, not proof of broken tracking. A direct-purchase path or the data-window boundary could also explain it.

Solutions considered

Option	Potential benefit	Trade-off
Investigate product discovery	Targets the larger observed stage gap	Views do not all imply intent; category and price mix can confound
Investigate cart-to-purchase	Focuses on stronger observed intent	No checkout, payment, stock, or shipping-fee event to identify cause
Audit instrumentation first	Clarifies missing categories and purchase chains	Slower path to a visible product change

The preferred first move is a short instrumentation audit, followed by a focused product investigation in one high-volume named category. The data does not justify a broad redesign or a revenue-uplift claim.

Conclusion

The largest observed drop is between view and cart, and named categories differ substantially. However, unknown categories and purchase journeys without a complete observed chain constrain diagnosis. This analysis narrows the next questions. It does not quantify lost revenue, establish causality, or measure checkout abandonment.

Next steps

Week 1 - Product analyst and analytics engineer: validate event definitions with the tracking owner, review a sample of unattributed purchase pairs, and determine whether a direct-purchase path exists. Set a category-label coverage target.

Week 2 - Product manager and UX researcher: choose one high-volume named category for qualitative review. Check product detail pages while collecting missing context on price, stock, and traffic source.

Weeks 3-4 - Product manager and experimentation analyst: preregister a narrow experiment only after the mechanism is understood. Define the eligible cohort, purchase conversion outcome, guardrails, exposure window, and decision rule before launch.

After the test - Analyst: report the observed treatment effect and uncertainty. If an experiment cannot run, present the work as prioritization rather than proven uplift.

Method and source

Python and SQLite process identifiers locally; only aggregate counts and charts appear here. Grain, denominators, and source-file hash are documented in the accompanying data dictionary. Source: REES46 / Open CDP, Kaggle electronics events. Raw events and user/session IDs are not part of the public-facing package.